KING SALMAN INTERNATIONAL UNIVERSITY

Field of ENGINEERING

Engineering Program
All Engineering Program

MAT121 Statics

3 Cr. Hrs. = (***2*** LCT + ***2*** TUT + ***0*** LAB + ***0*** OTH)

Lecture : (Vectors for mechanics)
Prof. Ayman Abdel hamed
Dr : Haytham Essam Amin

Date : 6 / 10 / 2026

Vector Representation:

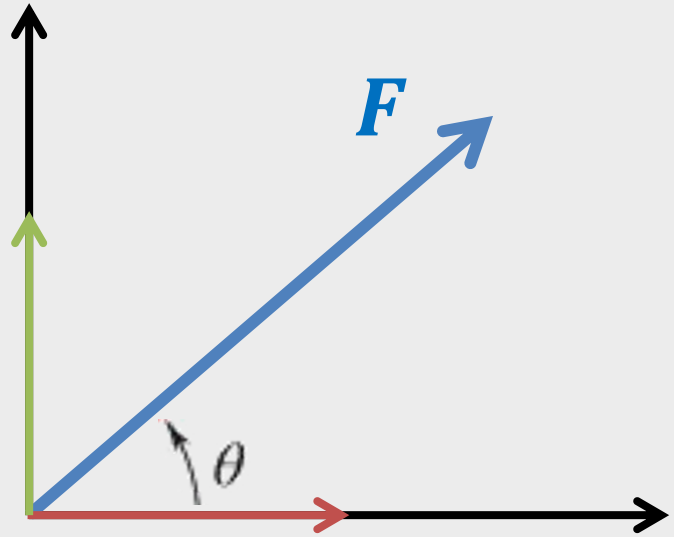
2- Analytical Representation

As shown the graphical presentation was a very simple method to represent vectors, also mathematical operations can be done easily. But as the number of vectors increase the mathematical operations will be tedious and need long time to be executed. So, the analytical representation is introduced.

Analytical Representation in Two Dimensions:

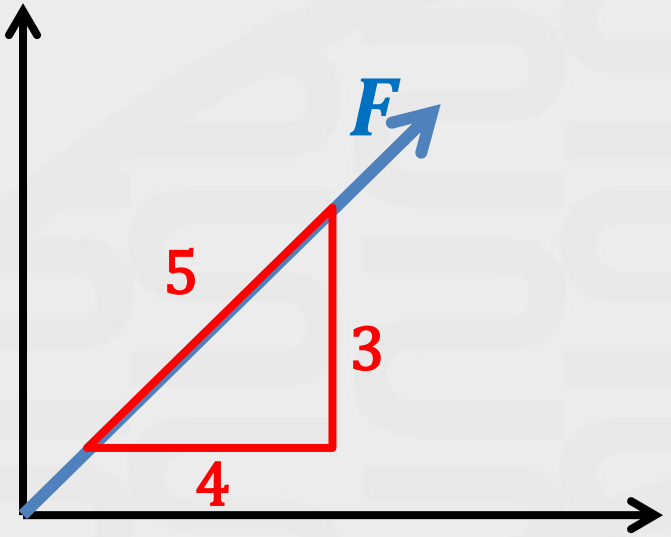
- When a vector is resolved into two components along the x and y axes, the components are then called *rectangular components*.
- In general, the components of an arbitrary vector $\vec{A}$ can be obtained by placing the tail of the vector at the origin, and by dropping perpendiculars from the head of the vector to the x and y axes.
- The intercepts of these perpendiculars with the axes (which may be positive or negative) give us the x and y components of the vector $\vec{A}$.

Rectangular Components of a Force



$$F_x = F\cos\theta$$

$$F_y = F\sin\theta$$



$$\vec{F}_x = \left(F \times \frac{4}{5}\right)$$

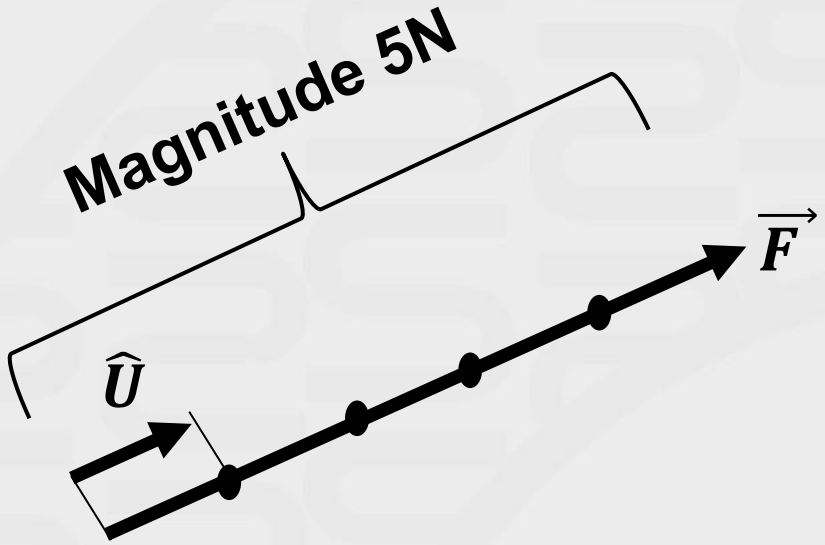
$$\vec{F}_y = \left(F \times \frac{3}{5}\right)$$

$$\boldsymbol{F} = \boldsymbol{F_x} + \boldsymbol{F_y}$$

Analytical Representation

Unit vector

A unit vector has magnitude one, with no units, means that if you multiply a unit vector by a scalar, the resulting vector has a magnitude equal to the value with-units of the scalar,



$$\vec{F} = 5 \ \hat{U}$$

$$\vec{F} = F \ \hat{U}$$

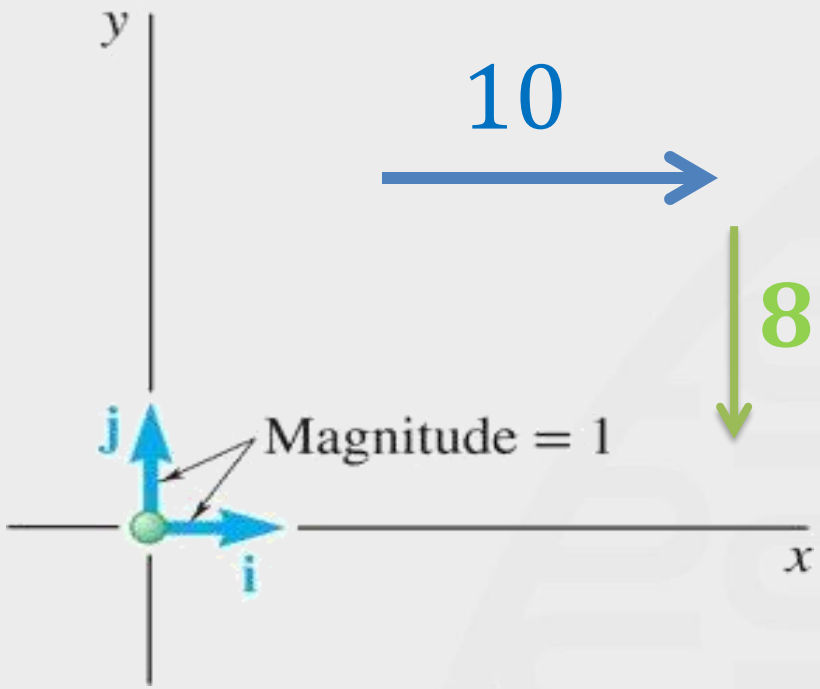
Vector

Magnitude

Direction

$$\hat{U} = \frac{\vec{F}}{F}$$

Cartesian Unit Vectors



Define perpendicular *unit vectors* $\boldsymbol{i}$ and $\boldsymbol{j}$ that are parallel to the *x* and *y* axes.

$$\vec{F} = 10\,\vec{i} \qquad \vec{F} = -8\,\vec{j}$$

Cartesian Coordinate System

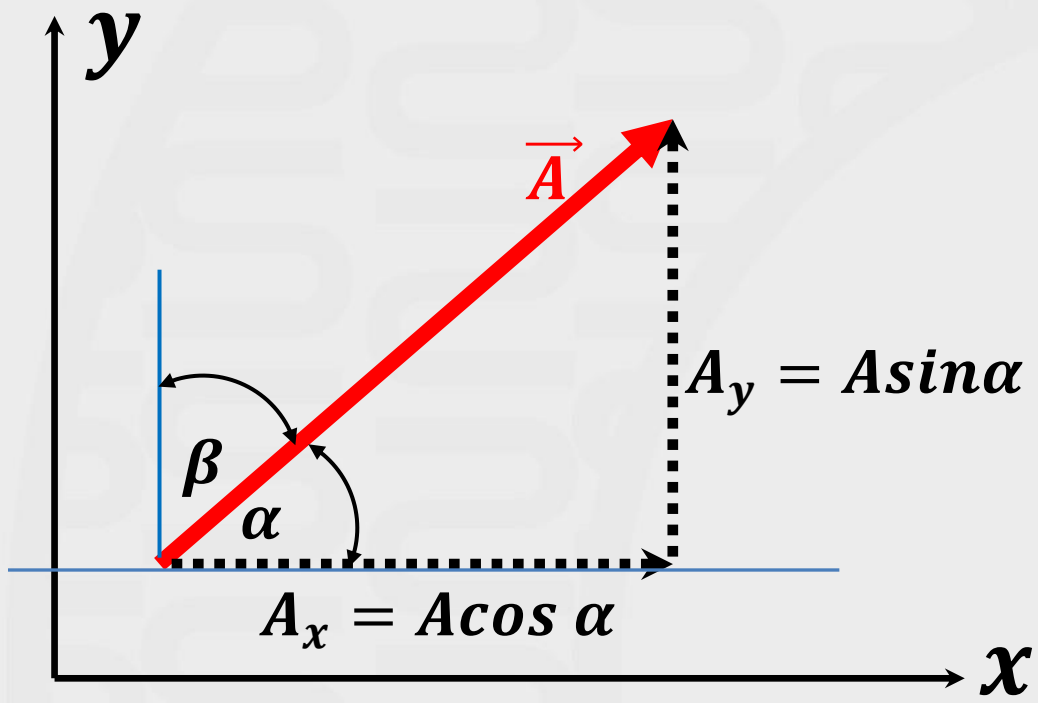
Vector Components Two-dimensions

Any vector in two dimensions can be broken down into two components along the x and y axes. Assume a vector $\vec{A}$ as shown in Figure is broken up or resolved into two components vectors.

$$\vec{A} = A\cos\alpha\, \boldsymbol{i} + A\sin\alpha\, \boldsymbol{j}$$

OR

$$\vec{A} = A\cos\alpha\, \boldsymbol{i} + A\cos\beta\, \boldsymbol{j}$$



4 STEPS For Solution



Cartesian Form

Convert each force to Cartesian form

$$\overrightarrow{\boldsymbol{F}} = F\cos\alpha\,\boldsymbol{i} + F\sin\alpha\,\boldsymbol{j}$$



Resultant Vector

i Components
j components



Resultant Magnitude

The magnitude of the vector

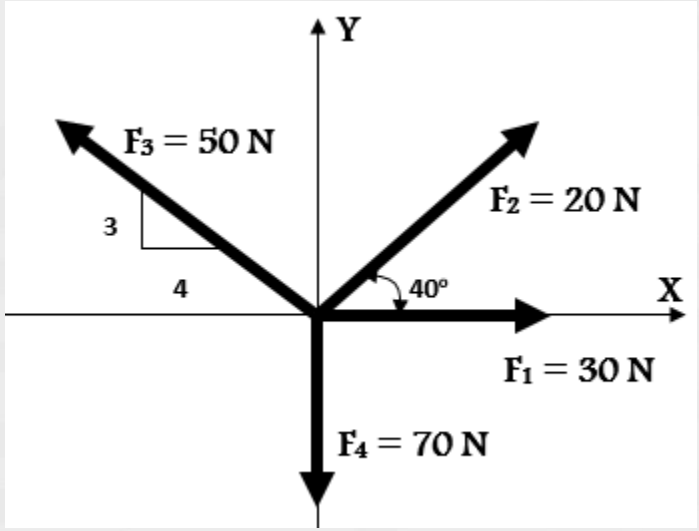
$$R = \sqrt{R_x^2 + R_y^2}$$



Resultant Dir.

$$\theta = \tan^{-1}\left|\frac{R_y}{R_x}\right|$$

Example 1 :



For the shown forces F1, F2, F3 and F4, determine the magnitude and direction of its resultant.

First, each force must be represented in Cartesian form to be added:

- For force F_1:

  Where the force F_1 coincides with x axis, then:

$$\overrightarrow{F_1} = 30\,\boldsymbol{i}$$

- For force F_2:

$$F_2 = 20\cos 40\;\boldsymbol{i} + 20\sin 40\;\boldsymbol{j}$$

$$\therefore \overrightarrow{F_2} = 15.32\,\boldsymbol{i} + 12.855\,\boldsymbol{j}$$

- For force F_3:

  Where the force has a slope triangle, then:

$$\overrightarrow{F_3} = -50 \times \frac{4}{5}\,\boldsymbol{i} + 50 \times \frac{3}{5}\,\boldsymbol{j}$$

$$\therefore \overrightarrow{F_3} = -40\,\boldsymbol{i} + 30\,\boldsymbol{j}$$

- For force F_4:

Where the force F_4 coincides with y axis, then:

$$F_4 = -70\,\boldsymbol{j}$$

Finally, add the forces by adding its components:

$$\therefore \vec{R} = \sum \vec{F_i} = (30 + 15.32 - 40)\vec{\imath} + (12.855 + 30 - 70)\vec{\jmath}$$

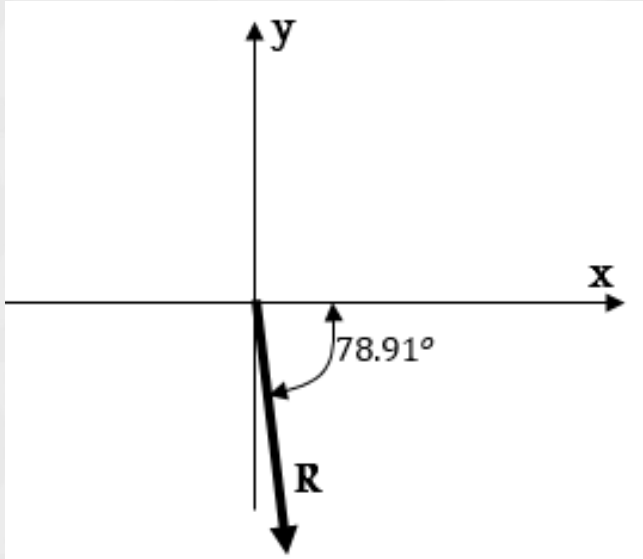
$$\therefore \vec{R} = 5.32\vec{\imath} - 27.145\,\vec{\jmath}$$

$$\therefore R = \sqrt{(5.32)^2 + (-27.145)^2}$$

$$= 27.661\ N$$

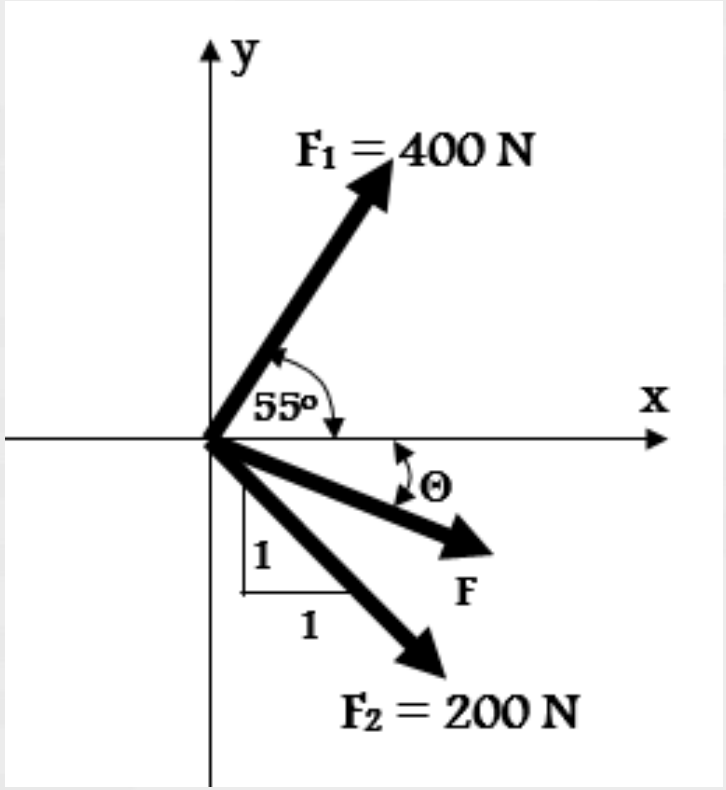
$$\& \tan\theta = \frac{-27.145}{5.32}$$

$$\therefore \theta = -78.91^o$$



Example 2:

If the resultant of the shown forces is to be 500 N and directed along positive x axis, determine the magnitude F and the angle Θ.



- For force F:

$$F_1 = F\cos\theta\ \boldsymbol{i} - F\sin\theta\ \boldsymbol{j}$$

- For force F_1:

$$\overrightarrow{F_1} = 400\cos 55\ \boldsymbol{i} + 400\sin 55\ \boldsymbol{j}$$

$$\therefore \overrightarrow{F_1} = 229.43\ \boldsymbol{i} + 327.66\ \boldsymbol{j}$$

- For force F_2:

$$\overrightarrow{F_2} = 200 \times \frac{1}{\sqrt{2}}\ \boldsymbol{i} - 200 \times \frac{1}{\sqrt{2}}\ \boldsymbol{j}$$

$$\therefore \overrightarrow{F_2} = 141.421\ \boldsymbol{i} - 141.421\ \boldsymbol{j}$$

$$\because \vec{R} = 500\,\vec{\imath} = \sum \vec{F_i}$$

$$= (F\cos\theta + 229.43 + 141.421)\vec{\imath}$$

$$+ \quad (-F\sin\theta + 327.66 - 141.421)\vec{\jmath}$$

$$\therefore 500\,\vec{\imath} = (F\cos\theta + 370.851)\vec{\imath} +$$

$$(-F\sin\theta + 186.239)\vec{\jmath}$$

By equating i th components and j th components, then:

For i th components:

$$\therefore 500 = F\cos\theta + 370.851$$

$$\therefore F\cos\theta = 129.149 \quad \Rightarrow A$$

For j th components:

$$\therefore 0 = -F\sin\theta + 186.239$$

$$\therefore F\sin\theta = 186.239 \quad \Rightarrow B$$

By dividing B with A, then:

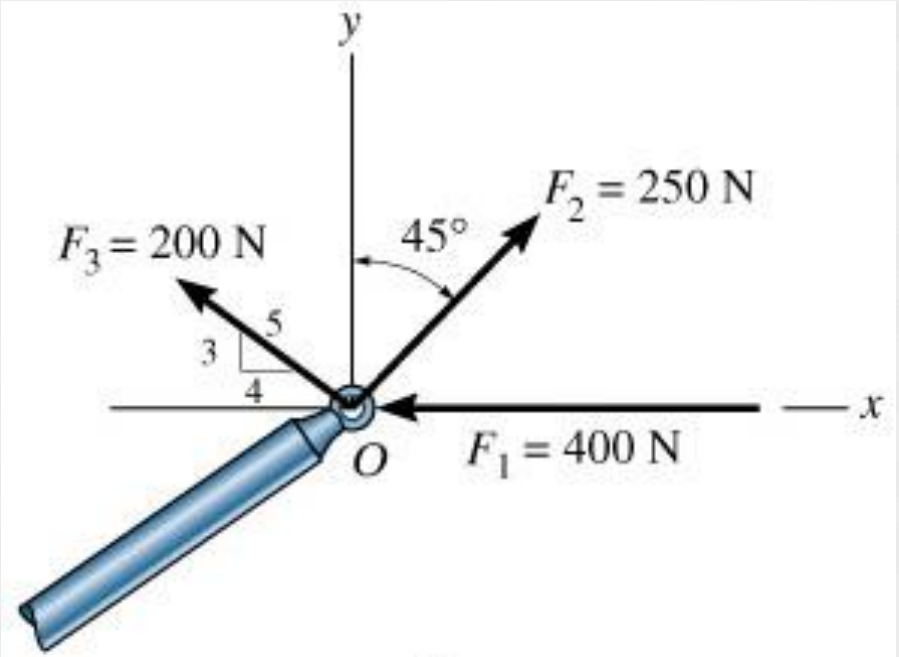
$$\therefore \tan\theta = 1.442$$

$$\therefore \theta = 55.26^o$$

$$\therefore F = 226.637\ N$$

Example 3

The end of the boom 0 in Figure. is subjected to three concurrent and coplanar forces. Determine the magnitude and direction of the resultant force.



SOLUTION:

Write the forces in terms of Cartesian unit vectors ***i*** and ***j***:

$$\vec{F}_1 = -400\,\boldsymbol{i}$$

$$\vec{F}_2 = 250\sin 45\,\boldsymbol{i} + 250\cos 45\,\boldsymbol{j} = \frac{250}{\sqrt{2}}\,\boldsymbol{i} - \frac{250}{\sqrt{2}}\,\boldsymbol{j}$$

$$\vec{F}_3 = -\left(200 \times \frac{4}{5}\right)\boldsymbol{i} + \left(200 \times \frac{3}{5}\right)\boldsymbol{j} = -160\,\boldsymbol{i} + 120\,\boldsymbol{j}$$

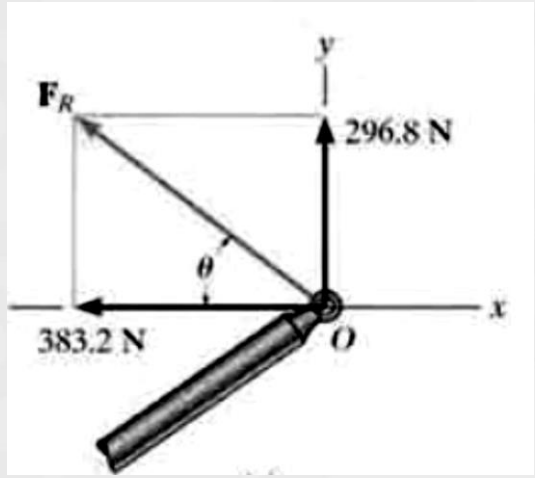
Example 2

The resultant force vector is

$$\vec{\boldsymbol{R}} = \sum \vec{\boldsymbol{F}} = \vec{\boldsymbol{F}}_1 + \vec{\boldsymbol{F}}_2 + \vec{\boldsymbol{F}}_3$$

$$\vec{\boldsymbol{R}} = \left(-400 + \frac{250}{\sqrt{2}} - 160\right)\boldsymbol{i} + \left(\frac{250}{\sqrt{2}} + 120\right)\boldsymbol{j}$$

$$\vec{\boldsymbol{R}} = -383.2\,\boldsymbol{i} + 296.8\,\boldsymbol{j}$$



The magnitude and direction of the resultant force

$$R = \sqrt{R_x^2 + R_y^2} = \sqrt{(-383.2)^2 + (296.8)^2} = 485\ N$$

$$\theta = \tan^{-1}\left|\frac{R_y}{R_x}\right| = \tan^{-1}\frac{296.8}{383.2} = 37.8°$$

KING
SALMAN
INTERNATIONAL UNIVERSITY

THANK YOU

16949 0 www.ksiu.edu.eg 0 info@ksiu.edu.eg

El Tur Campus: Sahely Road Ras suder Campus: Obour Street Sharm El Sheikh Campus: El Nour District